STOCKSCOPE

Architecture & Phase 1 User Flow

Slovenský študijný sprievodca • anglické názvy sekcií • príklady • argumenty na prezentáciu

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Tento materiál vysvetľuje schválenú dokumentáciu StockScope. Nie je to návrh jej zmien. Vysvetľujúce príklady a prezentačné formulácie sú označené ako študijný komentár. Úplné pôvodné anglické znenie je pripojené na konci ako referenčná príloha.

Hlavná myšlienka

StockScope nie je nový Warehouse Management System (WMS). Je to doplnková vrstva nad existujúcimi skladovými dátami, ktorá premieňa rozsiahly Location Report na prehľad o tom, čo sa v sklade nachádza, kde je problém, čo si vyžaduje kontrolu a ako sa situácia vyvíja v čase.

Ako tento dokument čítať

Časť I vysvetľuje zámer a tri fázy. Časť II prechádza používateľský tok a jednotlivé obrazovky. Časť III vysvetľuje najdôležitejšie pravidlá, import, históriu a architektúru dát. Časť IV pripravuje odpovede na otázky pri prezentácii. Časť V obsahuje úplné anglické znenie pôvodnej architektúry, aby nič z dohodnutého rozsahu nechýbalo.

Obsah

- I. Architecture, Development Roadmap, Business Analysis
- II. User Flow: Dashboard, Locations, Items, Push Items
- III. Status Classification, Review Workflow, Snapshots, Import Data
- IV. Presentation Notes, Questions & Answers, Study Checklist
- V. Original Architecture (English reference, full text)

ČASŤ I | ZÁMER A PLÁN

About StockScope

V sklade už existujú údaje o boxoch, nástrojoch, artikloch, statusoch a lokáciách. Problém nie je vždy nedostatok dát, ale čas a námaha potrebná na získanie praktickej odpovede. Zamestnanec často musí otvárať systém, filtrovať, exportovať Excel a manuálne spájať informácie.

StockScope má tento čas skrátiť. Namiesto ďalšieho evidenčného systému ponúkne **prehľad, upozornenia a súvislosti** nad dátami, ktoré už sklad používa.

> Prezentačná veta: „Nesnažím sa nahradiť existujúci systém. Chcem, aby dáta, ktoré už máme, pomáhali rýchlejšie rozhodovať.“

Čo StockScope nerobí: nemení pôvodné WMS statusy, nesľubuje skutočné potvrdenie transportu bez vhodného zdroja a v prvej fáze nevypočítava kompletné sety.

Architecture & Development Roadmap

Projekt má **tri produktové fázy** a samostatne **jedenásť architektonických krokov**. Tieto dve osi netreba zamieňať. Fázy hovoria, akú hodnotu aplikácia postupne prinesie; kroky hovoria, ako ju navrhneme a postavíme.

- **Phase 1 – Warehouse Visual:** Excel import, kontrola dát, prehľad skladu, Locations, Items, Reviews, Push Items, základné trendy.
- **Phase 2 – Warehouse Intelligence:** vzťahy medzi artiklami, úplné sety, nové kombinácie setov, pokročilá kontrola a jednoduché historické porovnávania.
- **Phase 3 – Advanced Capacity, History & AI:** fyzické rozmery, detailná história, Scan Report, hlbšia analytika a inteligentné návrhy.

Každá fáza má byť samostatne užitočná. To je dôležitý argument pri obhajobe rozsahu: výsledok Phase 1 nie je len technický prototyp, ale nástroj s reálnym denným použitím.

Phase 1 – Warehouse Visual

Vstup: existujúci Location Report v `.xlsx`. **Výstup:** aktuálny prehľad skladu a zoznam problémov, ktoré treba riešiť. **Zásada:** nečakať na finálnu fyzickú mapu skladu ani na kompletné rozmery lokácií.

Kľúčový dátový tok: **Location Report → Excel Import → Data Adapter → StockScope Models → Application**. Data Adapter prekladá štruktúru externého Excelu do stabilných interných modelov. Keď bude neskôr dostupné API, vymení sa adaptér, nie celá aplikácia.

Manuálna kapacita: lokácia `BS-F08` má limit 356 kusov a aktuálne 302. Obsadenosť je 302 / 356 = 84,8 % a voľných miest je 54. Ide o počet Itemov, nie o presný výpočet objemu. Bez spoľahlivého limitu sa percento nezobrazuje a nevznikajú kapacitné Reviews.

Prehľady lokácií: Phase 1 používa List View, teda tabuľku. Compact View a Map View sú budúce vizualizačné režimy nad rovnakými dátami. To umožní neskoršie rozšírenie bez duplicity logiky.

Základná Stock Control úloha: namiesto kontroly všetkých lokácií má človek dostať zoznam miest a Itemov, kde systém zistil výnimku.

Phase 2 – Warehouse Intelligence

Najdôležitejšia novinka je rozlišovanie medzi fyzickými Itemami a logickými **Article Relations**. Jeden kompletný set môže vyžadovať viacero boxov a nástrojov, ktoré ani nemusia ležať v jednej lokácii.

- **ALL:** vyžadujú sa všetky definované komponenty.
- **ANY:** postačuje jedna alternatíva zo skupiny.
- **ALL + ANY:** povinné položky a alternatívne skupiny sa môžu kombinovať.

Príklad: 10 vhodných boxov a 5 kompatibilných nástrojov znamená najviac 5 možných úplných setov, ak každý set potrebuje jeden box a jeden nástroj. Skenovanie však nemusí dokazovať, ktorý konkrétny

nástroj je v ktorom boxe, preto je dôležitý **výpočet z množstiev**.

„Find New Sets“ má odpovedať: ktoré kombinácie sa dajú vytvoriť, koľko ich je, čo chýba a v ktorých lokáciách ležia komponenty. Pokročilé Push pravidlá potom môžu pracovať aj s kompletnými setmi.

Phase 3 – Advanced Capacity, History & AI

Keď budú spoľahlivé rozmery lokácií a boxov, bude možné rozšíriť manuálnu kapacitu o fyzické výpočty podľa šírky, dĺžky, výšky a množstva. Dovtedy zostáva manuálny limit užitočným a transparentným riešením.

História zo snapshotov umožní sledovať zmeny stavov, presuny Itemov, rast zásob, často menené lokácie a dlhodobo neaktívne boxy. **Scan Report** môže doplniť čas skenu, používateľa, lokáciu, status a ďalšie udalosti, ak sú dostupné. Nie každý sken však nevyhnutne dokazuje fyzicky dokončený presun.

AI je až posledná vrstva. Základné pravidlá musia byť deterministické: napríklad obsadenosť 103 % je problém bez ohľadu na AI. AI môže až nad dôveryhodnými výsledkami navrhovať, kde sa oplatí konať.

Future Personal Project – Scanner UI Experiment

Toto je samostatný osobný experiment, nie súčasť Phase 1. Má skúmať rýchly scan-first pracovný tok: **SCAN → SEE INFORMATION → ACT**. Po naskenovaní H-kódu by používateľ okamžite videl identitu, status, lokáciu, navrhovaný cieľ a dostupné akcie.

Pri presune musí pracovník stále fyzicky naskenovať skutočnú cieľovú lokáciu. Návrh lokácie je iba pomôcka. Takýto projekt by mohol slúžiť ako samostatné Angular portfólio bez konkurencie voči existujúcemu firemnému skeneru.

Expected Improvements

Zmysel projektu sa dá obhájiť aj merateľne: menej času pri hľadaní H-kódu, pri kontrole lokácie, pri identifikovaní nesprávneho statusu, Push Itemu či obsadenosti skladu.

Praktický spôsob porovnania: zvoliť rovnakú úlohu, odmerať čas v súčasnom WMS/Excel postupe a v StockScope. Nie každá výhoda musí byť vyjadrená peniazmi, ale jednoduché časové porovnanie poskytne konkrétny dôkaz.

> Prezentačná veta: „Úspech nebudem merať tým, koľko obrazoviek vytvorím, ale tým, o koľko rýchlejšie človek nájde správnu odpoveď.“

Architecture Process

Jedenásť krokov dokumentácie: **1. Business Analysis, 2. User Flow, 3. Data Models, 4. Services & State, 5. Components, 6. Routing, 7. Implementation, 8. Validation, 9. Testing, 10. Data Flow Check, 11. Deployment & Production.**

Business Analysis a Phase 1 User Flow sú uzavreté. Ďalším krokom sú Data Models. Ostatné kapitoly sú v originálnej dokumentácii zatiaľ zámerne pripravené ako nadpisy; tento študijný sprievodca ich nevydáva za hotové rozhodnutia.

1. Business Analysis

Základné otázky projektu: Čo máme? Kde sa to nachádza? Čo je v konkrétnej lokácii? Aká je jej obsadenosť? Čo si vyžaduje pozornosť? Ktoré Push Items sú dostupné? Ako sa sklad mení?

V neskorších fázach pribudne: koľko úplných setov máme, ktoré nové sety sa dajú zostaviť, kde ležia komponenty, ktoré Itemy sa často presúvajú a ktoré sa dlho nehýbu.

Obhajoba Phase 1: poskytuje užitočné odpovede iba z dát, ktoré sú dostupné dnes. Nie je nutné čakať na zložité integrácie.

ČASŤ II | POUŽÍVATEĽSKÝ TOK

2. User Flow & Main Flow

Celý cyklus začína reportom: **Location Report → Import → Validation → Preview → Compare → Confirm Import → Update Warehouse State → Reconcile Reviews → Create Snapshot → Dashboard**.

Používateľ nikdy nemá aktualizovať produkčný prehľad už samotným výberom súboru. Najprv musí vidieť, čo sa importuje a ako sa nové čísla líšia od predchádzajúceho stavu.

Údaje z WMS a vlastné údaje StockScope musia zostať oddelené. Nový report aktualizuje stav skladu, ale nesmie vymazať poznámky, kapacity, Review History, posledné fyzické kontroly ani konfiguráciu Push Articles.

Dashboard

Dashboard je denná štartovacia obrazovka. Má ukazovať aktuálnu obsadenosť skladu, počet aktívnych Push Items, počet unikátnych Locations to Review, posledný import, najdôležitejšie problematické lokácie, Push Items a jednoduché trendy.

Zoznam Locations to Review ukáže približne osem prioritných lokácií. Tlačidlo View All otvorí Locations s prednastaveným filtrom a zoradením. Import Data musí byť ľahko dostupné.

Dôležité rozlíšenie: počet Locations to Review je počet rôznych lokácií, nie počet Review záznamov. Jedna lokácia môže mať tri Reviews, ale v počte lokácií je iba raz.

Locations to Review

Review je zistená podmienka, ktorú treba preveriť. Rozlišujeme **Location-level** a **Item-level**. Jedna lokácia môže mať súčasne kapacitný Review a niekoľko Item Reviews.

Location-level: Low Occupancy ≤15 %; Nearly Full ≥90 % a <100 %; Full =100 %; Over Capacity >100 %. Tieto kapacitné dôvody sú pre jednu hodnotu obsadenosti navzájom vylučujúce.

Item-level: Wrong Status; Unexpected / Mixed Item; Push Item. Item Review zostáva záznamom konkrétneho H-kódu, ale jeho lokácia sa zobrazí v zozname lokácií na kontrolu.

Priority od najnižšej po najvyššiu: Low Occupancy, Nearly Full, Unexpected / Mixed Item, Wrong Status, Push Item, Full, Over Capacity. Pri triedení lokácia zdedí najvyššiu prioritu zo svojich aktívnych Reviews.

Príklad: lokácia má 103 % obsadenosť a dva Itemy s nesprávnym statusom. V Locations to Review je započítaná raz, ale do celkového počtu aktívnych Reviews prispeje tromi záznamami.

Locations

Jedna centrálna stránka `/locations` slúži pre všetky prehľady. Filtre: All, To Review, Handled, Review Reason, rozsah obsadenosti, vyhľadávanie podľa kódu a triedenie.

Filtre a triedenie majú byť uložené v URL query parametroch, napríklad `/locations?status=review&sort=priority`. Odkaz sa dá zdieľať a po návrate z detailu sa nestratí pracovný kontext.

Riadok lokácie ukáže kód, obsadenosť, počet Itemov, aktívne Reviews a Last Checked. **Last Checked** je fyzická kontrola pracovníkom, nie čas posledného importu.

Pri filtroch sa zobrazujú počty zodpovedajúcich lokácií. Dôvody bez zhody môžu zostať viditeľné, ale deaktivované. Pokročilé vyhľadávanie podľa Article/H-kódu je až budúce rozšírenie.

Location Detail

Detail lokácie spája prehľad lokácie, Location Reviews, zoznam všetkých Itemov, filtre a krátku históriu. V Phase 1 sa zobrazuje celý dostupný zoznam Itemov v lokácii, nie iba náhľad.

V hlavičke: kód, obsadenosť, počet Itemov, počet aktívnych Reviews, prítomné Articles s množstvami, Last Checked a kapacita, ak je známa.

Location Reviews sú oddelené od Item Reviews. Item-level dôvod sa zobrazuje pri konkrétnom H-kóde. Každý Review možno riešiť samostatne.

Zoznam Itemov podporuje vyhľadávanie podľa H-kódu, Article, Status a režim All / Reviews only. Viacero Article filtrov funguje spoločne. Príklad: `Showing 7 of 22 items`.

Mark as Checked zaznamenáva fyzickú kontrolu a aktualizuje Last Checked. **Mark as Handled** je úkon nad konkrétnym Review. Jedno neznamená druhé.

História lokácie môže ukázať trend obsadenosti, napríklad 76 → 82 → 94 → 103 %, a nedávne udalosti. Pri návrate na Locations sa majú zachovať filtre a triedenie.

Item Detail

Kliknutie na H-kód otvorí **modal**, nie samostatnú stránku. To platí na Items Page aj na Location Detail. Používateľ nestráca scroll, filtre ani kontext.

Informácie: H-code, Article, Customer, Customer Article Number, Location, Level, Status, rozmery, Call Off, aktívne Reviews a Article Relations, keď existujú spoľahlivé dáta.

`kinds` je vstupné pole, z ktorého sa môže odvodiť zrozumiteľné Call Off. Napríklad `35 Call Off List ASML`, `RTM - Supplier Network` sa používateľovi zobrazí ako `Call Off: ASML`. Rozmery sú v Phase 1 informatívne.

Relations: pôvodné HQ dáta môžu obsahovať `articleSet`, ale Location Report nemusí poskytovať dostatočné podklady. Ak nie sú spoľahlivé, sekcia sa nezobrazuje. ALL/ANY a kompletné sety patria do Phase 2.

Item Notes: jedna editovateľná poznámka na jeden H-kód, uložená so zmenovým časom. Patrí Itemu, nie Article. Import ju nikdy nesmie vymazať; zostáva zachovaná aj keď H-kód dočasne zmizne.

Poznámka nemení WMS status, nevytvára Review, neoznačuje ho za Handled ani ho nerieši. Príklad: Item `On hold`, vysvetlenie „Čakáme na odpoveď zákazníka; potvrdené supervisorom.“

Article Image: prípadný obrázok patrí Article a môže sa používať pre všetky jeho H-kódy. V Phase 1 nie je povinný.

Items

Globálna stránka Items zobrazuje jeden riadok na jeden H-kód. Je vhodná, keď poznáš Item alebo Article, ale nechceš najprv prechádzať cez lokáciu.

Vyhľadávanie podporuje H-code, Article a Customer Article Number. Filtre zahŕňajú Article, Location, Status, Review, Size a Call Off.

Logika filtrov: viacero hodnôt v jednej skupine sa kombinuje cez OR; rôzne skupiny cez AND. Príklad: `(ZEISS021 OR ZEISS022) AND (BS-A01 OR BS-A04) AND Storage AND Large`.

Size je vlastnosť Article. Small / Medium / Large majú vzniknúť až po analýze reálnych rozmerov, nie podľa ľubovoľne vymyslených hraníc.

Call Off je tiež vlastnosť Article. V tabuľke môže byť malá ikonka s vysvetlením pri hoveri, nie nutne ďalší široký stĺpec.

Kliknutie na Item otvorí rovnaký Item Detail modal. Kliknutie na Location otvorí detail lokácie. URL query parametre umožnia napríklad `/items?location=BS-A04&article=NEWAYS56`.

Push Items & Push Articles

Push Item je individuálny H-kód, ktorého Article je označený v StockScope ako Push Article a ktorý podľa príslušného workflow ešte vyžaduje akciu. **Konfigurácia patrí Article; Review patrí Itemu.**

V Phase 1 existujú dva workflow typy: **ASML Push** s každodenným harmonogramom a očakávaným statusom `Transported to ASML`; **Outbound Push** s voliteľným dňom v týždni a statusom `Transported to customer`.

Oba používajú očakávanú externú lokáciu `EXTERN-NL-EIN-01`. Táto lokácia znamená externý odovzdávací krok, **nie dôkaz fyzického doručenia**.

Konfigurácia Push Article: Article, workflow typ, očakávaný deň, voliteľná poznámka, dátum poslednej zmeny. Spravuje sa cez modal na Items Page. Nie je potrebná samostatná Push Items stránka.

Aktívne Push Reviews sa zobrazujú filtrom `/items?review=push-item`. Dashboard používa rovnakú definíciu počtu. `Sending Soon` a `Expected Today` sú iba informačné indikátory, nie nové Review Reasons.

Zásadné pravidlo: keď aktívny Push Item H-kód chýba v nasledujúcom úspešne potvrdenom Location Report, jeho Review sa automaticky vyrieši s dôvodom `No longer present in monitored warehouse`. To nedokazuje doručenie ani konkrétny transportný status.

Ak sa rovnaký H-kód neskôr vráti a stále spĺňa Push pravidlá, vznikne nový Review; stará história zostane zachovaná. Phase 2 môže rozšíriť Push pravidlá o vzťahy medzi Articles a kompletné sety.

ČASŤ III | PRAVIDLÁ A SPOĽAHLIVOSŤ

Item Status Classification

StockScope číta **pôvodný WMS status bez zmeny**. Status Classification hovorí, ako ho interpretovať a zobraziť. Review Detection samostatne rozhoduje, či vznikne problém na preverenie.

Normal Storage Statuses: `Storage`, `Storage - Cleanroom clean`, `RTM- New Build`, `Storage - Reserved for ASML Factory`. Ide o platné skladové stavy. Storage má modrý indikátor, Cleanroom clean ružový, RTM New Build červený, ale červená v tomto prípade **neznamená chybu**.

Damage and Repair Workflow: `Ready for Repair` (poškodený prázdny box) a `Ready for Inspection at Tech Repair` (poškodený tool alebo box s toolom) sú platné **iba v** `Transfer to Run 2`. Na inej lokácii vytvoria Wrong Status Review.

Transport Statuses: `Transported to ASML` a `Transported to customer` sú platné iba pri `EXTERN-NL-EIN-01`. Ak je Item so statusom Transported stále v `BS-A03`, vzniká Wrong Status Review. Item správne presunutý do externej lokácie z Location Report zmizne.

Wrong Status Reviews: `Cleanroom Cleaning`, `Industrial Cleaning`, `Ready for Cleanroom Cleaning`, `Ready for Industrial Cleaning`, `Ready for Storage`, `Ready for Transport`, `Industrially cleaned`, `Stock Correction`. Ak sú tieto stavy stále v monitorovanom sklade, vzniká Wrong Status Review.

Unexpected / Mixed Item Reviews: `Inspected - Waiting for Tech Repair Parts`, `Intern repair CE`, `On hold`, `Pending PE`, `Waiting for Parts`, `Tech Repairing`, `Scrapped`. Môžu byť prevádzkovo legitímne, ale vyžadujú vysvetlenie alebo kontrolu.

Attention Indicators: doplnkový vizuálny signál, napríklad oranžový, nie ďalší Review Reason. Jedna podmienka nesmie vytvárať duplicitné Reviews. Neznáme statusy sa majú zobraziť, nie automaticky označiť za zdravé alebo nesprávne.

> Študijný príklad: `Ready for Repair` v `Transfer to Run 2` je správne. Presne ten istý status v `BS-A03` je nesprávne. Dôležitý nie je iba text statusu, ale kombinácia status + lokácia.

Review Workflow

Základný životný cyklus: **OPEN → Mark as Handled → WAITING FOR VERIFICATION → RESOLVED**. Označenie Handled znamená, že používateľ vykonal akciu, **nie** že StockScope overil výsledok.

Po ďalšom potvrdenom importe sa podmienka vyhodnotí znova. Ak zmizla, Review prejde na RESOLVED. Ak pretrváva, vráti sa z WAITING FOR VERIFICATION na OPEN.

Existuje aj **OPEN → RESOLVED**, keď sa problém odstráni bežnou skladovou aktivitou bez zásahu používateľa v StockScope. Príklad: nízka obsadenosť zmizne po doplnení lokácie.

Review History uchová detekciu, Handled, výsledok overenia a prípadný opätovný výskyt. Vyriešené Reviews už nezvyšujú aktuálny zoznam, ale historicky nezmiznú.

Pri Push Itemoch je potvrdená neprítomnosť H-kódu v celom Best Location Report dostatočná na uzavretie Review s dôvodom „No longer present in monitored warehouse“, nie na tvrdenie o doručení.

Snapshots & Trends

Každý úspešne potvrdený import uchová tri vrstvy histórie: **Warehouse Snapshot Summary**, **Location Snapshots** a **Historical Import Data**. Všetky patria k tomu istému importu a neskoršie importy ich neprepisujú.

Warehouse Snapshot Summary: čas, celková obsadenosť, počet Itemov, počet aktívnych Reviews a počty podľa dôvodu. Location Snapshots: kód lokácie, obsadenosť, počet Itemov a aktívnych Reviews. Historical Import Data: jednotlivé H-kódy, Articles, lokácie, levely a statusy.

Pôvodný Excel je vhodné archivovať, ak to technické a prevádzkové pravidlá dovoľujú. Neúspešný, zrušený alebo nepotvrdený import nevytvára historické záznamy.

Za deň môže existovať viac snapshotov (napr. 08:00, 12:00, 16:00). Pre jednoduchý denný graf sa použije najnovší dnešný snapshot a posledný snapshot predchádzajúcich dní.

Trends v Phase 1: Warehouse Occupancy, Total Items, Reviews. Predvolené intervaly môžu byť 7, 30 a 90 dní. Reviews možno filtrovať podľa Review Reason.

Historický snapshot sa **nesmie spätne prepočítať** po vyriešení Review. Ak v pondelok bolo 14 Push Reviews a v utorok 9, pondelková hodnota zostane 14. Inak by história prestala opisovať minulosť.

Import Data

Import je najcitlivejšia časť Phase 1, pretože zlý súbor by mohol zmeniť všetky počty, uzavrieť množstvo Reviews a poškodiť históriu. Preto je postup zámerne viacstupňový.

Select File: vyberie sa `.xlsx` Location Report. **File Validation:** súbor musí byť čitateľný, správneho typu, s očakávanou štruktúrou a povinnými stĺpcami. Chýbajúci povinný stĺpec je Error a blokuje potvrdenie.

Import Preview: zobrazí porovnanie starého a nového stavu, napríklad počet lokácií 184 → 184, Items 4 345 → 4 382, obsadenosť 66 % → 68 %, Reviews 27 → 21. Preview zatiaľ nič neukladá.

Import Warnings: dramatický pokles počtu Itemov, takmer nulová obsadenosť či veľká zmena počtu lokácií vyvolajú upozornenie. Technicky platný report môže mať aj legitímne veľké zmeny, preto warning automaticky neblokuje import.

Error vs Warning: Error = nedá sa spoľahlivo spracovať, import stop. Warning = spracovanie je možné, ale používateľ má overiť, či vybral správny report; potom môže potvrdiť alebo nahrať iný.

Confirm Import: až tu sa zmení aktuálny skladový stav, prepočítajú Reviews a uložia snapshoty. Operácia musí byť konzistentná: buď sa úspešne vykonajú všetky časti, alebo zostane predchádzajúci potvrdený stav. Opakovaný import rovnakého reportu sa má rozpoznať.

Review Reconciliation: nový report môže vytvoriť Review, ponechať ho OPEN, vrátiť Handled do OPEN alebo ho vyriešiť. StockScope-owned dáta, poznámky, konfigurácie a história zostávajú.

Missing Item Verification: Location Report pokrýva celý sklad Best. Ak Push H-kód v ďalšom potvrdenom reporte chýba, jeho Review sa automaticky uzavrie. Dôvod je „No longer present in monitored warehouse“. Aplikácia netvrdí, že Item bol doručený zákazníkovi.

Import Complete: po úspechu sa zobrazí zhrnutie nového stavu a rozdielov oproti predchádzajúcemu importu.

> Študijný príklad: omylom exportovaný report so všetkými nulami a zachovanými stĺpcami môže prejsť štrukturálnou kontrolou. StockScope preto porovná počty a ukáže výrazné Warning. Používateľ má možnosť nepokračovať.

Phase 1 Navigation & Phase 1 User Flow

Bočné menu Phase 1: **Dashboard, Locations, Items, Trends, Import Data**. Push Items sa otvárajú filtrom na Items, nie samostatnou stránkou.

Bežný deň: nový report → kontrola preview → potvrdenie → Dashboard → Locations to Review → detail lokácie → Item Detail → vysvetlenie v poznámke alebo Mark as Handled → ďalší import → automatické overenie.

Iný scenár: Dashboard → Push Items → filtrovaná Items Page → konkrétny H-kód. Ďalší scenár: Dashboard → Trends → zmeny obsadenosti alebo Review počtov.

Pravidlo dobrého UX: používateľ nemá stratiť kontext. Modaly sa zatvárajú na pôvodnú stránku, query parametre uchovávajú filtre a navigácia nevyžaduje opakované nastavovanie.

3. Data Models to 11. Deployment & Production

V schválenej architektúre sú tieto kapitoly pripravené na ďalšiu prácu. Nie sú ešte rozpracované ako finálna technická špecifikácia. Výnimkou je stručná sekcia **UI Design System** v kapitole Components.

Z dokumentácie už vyplývajú budúce entity a hranice: Article vs Item, Location, Review, Push Article Configuration, Item Note, Import, Snapshot a historické záznamy. Presné TypeScript rozhrania, databázové tabuľky a služby sa však budú rozhodovať až v ďalších krokoch.

UI Design System: spoločné pravidlá pre spacing, typografiu, farby, status indikátory, tlačidlá, karty, tabuľky, bloky lokácií, upozornenia a responzívne rozloženie. Reusable Angular komponenty a design tokens majú prednosť pred opakovaním CSS.

Pri prezentácii je férové povedať: „Business pravidlá a používateľský tok sú navrhnuté. Implementačná architektúra, dátové modely a testy sú ďalším krokom.“

ČASŤ IV | PRÍPRAVA NA PREZENTÁCIU

Presentation Notes – 30-second Pitch

> „StockScope je doplnková aplikácia pre sklad, ktorá mení existujúci Location Report na praktický prehľad. Ukáže, kde je sklad plný, ktoré lokácie potrebujú kontrolu a ktoré Push Items čakajú na akciu. Nechcem nahrádzať súčasný systém. Chcem využiť dáta, ktoré už máme, aby Stock Control vedel rýchlejšie nájsť to podstatné. Prvú verziu viem postaviť na Excel importe, bez zásahu do existujúceho WMS.“

Presentation Notes – 2-minute Walkthrough

1. Problém: sklad má veľa informácií, ale každodenné odpovede vyžadujú manuálne filtrovanie a exporty.

2. Riešenie: Location Report sa bezpečne importuje a prevedie na interné modely. Dashboard ukáže prehľad a výnimky.

3. Konkrétny príklad: otvorím lokáciu s 103 % obsadenosťou, vidím problémové H-kódy, ich statusy a môžem zaznamenať, čo bolo skontrolované.

4. Push workflow: viem nájsť všetky aktívne Push Items a ich očakávané odoslanie. Keď H-kód zmizne z nasledujúceho kompletného reportu, Review sa uzavrie bez falošného tvrdenia o doručení.

5. Bezpečnosť dát: import má validáciu, porovnanie, potvrdenie a historické snapshoty. StockScope nemení originálne WMS statusy.

6. Budúcnosť: vzťahy a sety v Phase 2, detailná história a kapacita v Phase 3. Najprv však doručiť malú funkčnú hodnotu.

Questions & Answers

Prečo nestačí Excel?

Excel je dobrý zdroj, ale opakované manuálne filtrovanie a porovnávanie zaberá čas. StockScope nad tým vytvorí jednotné pravidlá, uložené poznámky, Review workflow, históriu a prepojené obrazovky.

Prečo nie priamo API?

API môže byť výhodné neskôr, ale Phase 1 nemá byť blokovaná prístupovými právami alebo integráciou. Data Adapter oddelí formát zdroja od aplikácie, takže budúca zmena vstupu bude jednoduchšia.

Ako zabrániš poškodeniu dát?

Import sa neaplikuje pri výbere súboru. Najprv validácia, preview a porovnanie. Chyby blokujú, podozrivé zmeny varujú, potvrdenie je vedomý krok. Aktualizácia stavu, Reviews a histórie má byť jedna konzistentná operácia.

Čo znamená Review?

Konkrétny zistený problém alebo výnimka podľa obchodného pravidla. Je odlíšený od fyzickej kontroly lokácie a od poznámky používateľa.

Znamená Handled vyriešené?

Nie. Handled znamená, že človek vykonal akciu. RESOLVED vzniká až keď novšie dáta potvrdia, že podmienka už neexistuje.

Ako vieš, že Push Item odišiel?

V Phase 1 viem len to, že už nie je v nasledujúcom potvrdenom kompletnom Location Report pre Best. Neviem z toho potvrdiť skutočné doručenie. Na to treba ďalší dôveryhodný zdroj, napríklad budúci Scan Report.

Čo ak report obsahuje všetky nuly?

Aj technicky správny súbor môže byť obsahovo nezmyselný. Preto porovnávame nové hodnoty s predchádzajúcim stavom a zobrazujeme upozornenia pred potvrdením.

Prečo nie kompletné sety už v prvej fáze?

Vyžadujú spoľahlivé Article Relations a kombinácie ALL/ANY. Phase 1 vie priniesť hodnotu aj bez toho. Rozdelenie znižuje technické riziko.

Čo bude prvý merateľný výsledok?

Porovnanie času potrebného na vyhľadanie Itemu, kontrolu lokácie, nájdenie nesprávneho statusu alebo Push Itemu v starom a novom postupe.

Ako sa odlíši Article od Item?

Article opisuje typ produktu, spoločné vlastnosti a konfiguráciu. Item je konkrétny fyzický H-kód so svojou lokáciou, statusom, poznámkou a Review históriou.

Je to už hotový produkčný systém?

Nie. Uzavreté sú Business Analysis a Phase 1 User Flow. Data Models, služby, implementácia, testovanie a deployment sú nasledujúce architektonické kroky.

Key Talking Points

- **Complement, not replacement:** žiadne súperenie s existujúcim WMS.
- **See what matters:** systém upozorní na výnimky namiesto zahltenia všetkými riadkami.
- **Start with existing data:** Phase 1 funguje aj bez nového API.
- **Trust before automation:** validácia a potvrdenie pred zmenou skladu.
- **History from day one:** každým potvrdeným importom rastie hodnota historických dát.
- **Clear boundaries:** Handled nie je Resolved; zmiznutie Itemu nie je potvrdené doručenie.
- **Incremental value:** každá fáza prináša samostatný praktický výsledok.

Study Checklist

Pred prezentáciou by si mal vedieť vlastnými slovami vysvetliť tieto otázky:

- Prečo StockScope existuje a prečo nie je nový WMS?
- Čo presne je Phase 1 a čo zámerne odkladáme?
- Prečo Data Adapter a aký je rozdiel medzi Excel importom a budúcim API?
- Aký je rozdiel medzi Article a H-code Item?
- Ako sa počíta obsadenosť a čo keď kapacita nie je známa?
- Prečo sa počet Locations to Review nerovná počtu Reviews?
- Ako fungujú Wrong Status, Unexpected / Mixed Item a Push Item?
- Aký je rozdiel medzi Mark as Checked, Mark as Handled a Resolved?
- Prečo Push Item bez H-kódu v novom reporte neznamená potvrdené doručenie?
- Čo chráni aplikáciu pred chybným Excelom?
- Čo sa ukladá do snapshotov a prečo sa minulosť neprepisuje?
- Ako by si demonštroval, že StockScope skutočne šetrí čas?

Záver študijného sprievodcu

Najsilnejší argument StockScope je jednoduchý: **dáta už máme; teraz ich chceme používať múdrejšie, rýchlejšie a bezpečne.** Prvá fáza nie je o AI ani o komplikovanej automatizácii. Je o spoľahlivom prehľade, zrozumiteľných pravidlách a jasnom pracovnom toku.

ČASŤ V | ORIGINAL ARCHITECTURE (ENGLISH REFERENCE)

Nasleduje úplný text finálnej anglickej dokumentácie, priložený bez obsahových úprav ako referenčný zdroj. Táto časť zabezpečuje, že všetky konkrétne formulácie, príklady a detaily z originálu sú v PDF k dispozícii aj pri neskoršom štúdiu.

About StockScope

I work with warehouse stock every day and often see the same problem:

the data already exists, but getting a simple and useful answer from it can take too much time.

The goal of StockScope is not to replace, duplicate or compete with the existing warehouse management system.

StockScope works as an additional layer on top of the warehouse data that already exists.

Its purpose is to make that data easier to see, understand and use, and to help turn it into useful operational information.

Architecture

Development Roadmap

StockScope will be developed in three main phases.

Phase 1 – Warehouse Visual

Phase 2 – Warehouse Intelligence

Phase 3 – Advanced Capacity, History & AI

Each phase should produce something useful on its own.

Phase 1 – Warehouse Visual

Goal

The first version should turn the existing Location Report into a clear visual overview of the warehouse.

The warehouse layout is still changing, so StockScope should not depend on having final location dimensions or a final warehouse map before development can start.

Data Import

The first version can use the existing Location Report.

Data flow:

Location Report → Excel Import → Data Adapter → StockScope Models → Application

StockScope should work with its own internal models.

If API access becomes available later, Excel can be replaced by an API adapter without rebuilding the main application.

Future flow:

HQ API → API Adapter → StockScope Models → Application

Data Snapshots

StockScope should start preserving warehouse snapshots as early as possible.

Each imported report can represent the state of the warehouse at a specific point in time.

These snapshots do not need to provide advanced historical analytics in Phase 1.

The purpose is to start building historical data that can be used by later versions of StockScope.

Possible information to preserve includes:

- import date and time
- total number of objects
- objects per location
- article quantities
- statuses
- location occupancy
- Push Items

Starting this early prevents useful historical information from being lost before the historical analysis features are developed.

Historical Data Strategy

Each successfully confirmed Location Report import should preserve historical information at three levels:

- Warehouse Snapshot Summary
- Location Snapshots
- Historical Import Data

The Warehouse Snapshot Summary contains:

- import timestamp
- overall warehouse occupancy
- total number of Items
- total number of active Reviews

- active Review counts by Review Reason

Location Snapshots preserve historical information for individual warehouse locations, including:

- location code
- occupancy
- item count
- active Review count

Historical Import Data preserves the warehouse records associated with the confirmed import, including Item H-codes, Articles, locations, levels and statuses.

The original confirmed Excel report should also be archived when technically and operationally permitted.

Historical records must remain independent from the current warehouse state and must not be overwritten by later imports.

All historical records created by one import should be associated with the same confirmed import.

Phase 1 uses Warehouse Snapshot Summaries and Location Snapshots for basic historical trends.

Detailed Item movement analysis, location turnover and advanced historical comparisons belong to later phases.

Failed, cancelled or unconfirmed imports must not create historical records.

Warehouse Overview

The Locations page provides a unified overview of warehouse locations.

In Phase 1, the Locations page uses a table-based List View.

The List View allows users to:

- search for locations
- filter locations by review reasons
- filter by occupancy range
- sort locations
- identify locations requiring attention
- open Location Detail

The Locations page also supports predefined filters when opened from the Dashboard or other application areas.

Future Visualization Modes

In later phases, the Locations page can support multiple visualization modes:

- List View
- Compact View
- Map View

List View

The standard table-based overview of warehouse locations introduced in Phase 1.

Compact View

A compact visual representation where locations are displayed as regularly arranged blocks.

Each block represents a warehouse location and can use colors or indicators to communicate occupancy, reviews and other important conditions.

Map View

A visual representation based on the actual physical warehouse layout.

Locations are positioned according to the warehouse map, allowing users to understand their physical arrangement and identify areas requiring attention.

All visualization modes should use the same underlying location data, filtering logic and review information.

Switching between views should not require separate location management systems.

Selecting a location from any visualization mode should open the same Location Detail.

Compact View and Map View are future enhancements and are not required for Phase 1.

Location Detail

A single warehouse location can contain multiple objects and multiple article types.

This is especially important for locations containing tools, where several different articles may be stored in the same location.

A location detail should therefore be able to show multiple stock records, including:

- object / H-code
- article
- status
- active reviews
- other relevant stock information

This makes it possible to move from seeing that a location has a problem to seeing exactly which objects or articles are causing it.

Manual Capacity

Until reliable location dimensions are available, maximum capacity can be configured manually.

For example:

BS-F08 → Maximum 356

If 302 objects are currently stored there, StockScope can calculate:

302 / 356 → 84.8% occupied → 54 available

The data model can already contain dimensions for future use even if they are not yet used in the calculation.

If a location does not have a reliable configured capacity, its occupancy should be displayed as unavailable rather than assumed to be 0%.

Capacity-based Reviews must not be generated for locations without a valid capacity.

These locations should remain visible in StockScope and can still contain Items and Item Reviews.

Status Visualization

Important or unusual statuses should be visible directly from the warehouse overview.

A location can show a small visual indicator when something inside requires attention.

Opening the location then shows the exact object and its status.

The purpose is to make exceptions easy to notice without filling the overview with unnecessary information.

Push Items

Push Items are identified through StockScope-owned Push Article configuration.

In Phase 1, the user manages a list of Articles that should be treated as Push Articles.

Each Push Article configuration contains:

- Article
- Push workflow type (ASML or Outbound)
- Expected sending day
- Note (optional)
- Last changed date (automatic)

The sending schedule depends on the workflow type:

- ASML Push: Every day
- Outbound Push: Configurable day of the week

Both workflows use `EXTERN-NL-EIN-01` as their expected final location.

Their expected final statuses differ:

- ASML Push: `Transported to ASML`
- Outbound Push: `Transported to customer`

StockScope uses current warehouse data to find individual H-code Items belonging to configured Push Articles.

Matching Items are evaluated against their Push workflow rules and receive an active Push Item Review when further action is required.

Push Item Reviews are derived automatically and are not manually assigned to individual H-codes.

Transferred Items disappear from the available Location Report. Phase 1 therefore cannot directly verify their final transport status or destination.

An active Push Item Review is automatically resolved when its Item is absent from the next successfully confirmed Location Report.

The resolution reason must indicate that the Item is no longer present in the monitored warehouse, not that its final transport was verified.

Push Articles are managed through a modal on the Items page.

Active Push Items are displayed using the Items page with a predefined filter, rather than a separate Push Items page.

Push Article configuration is stored independently from imported Location Report data and must be preserved during future imports.

Expected Sending Day indicators provide operational context without creating additional Review Reasons.

In Phase 2, Push Item detection can be extended using Article Relations and Set Rules, including combinations of Articles required to form complete sets.

Basic Stock Control

Phase 1 should already provide a simple Stock Control view based on information available from the imported warehouse data.

It should help identify:

- unexpected statuses
- locations requiring attention
- available Push Items
- other basic stock exceptions that can be detected directly from the current data

The purpose is to help the stock controller focus on exceptions instead of manually checking everything.

More advanced Stock Control functions that depend on relationships between boxes and tools will be added in Phase 2.

Phase 2 – Warehouse Intelligence

Goal

Phase 2 should move StockScope from simply displaying warehouse data to understanding relationships inside that data.

Complete Sets

StockScope should understand which boxes and tools create a complete set.

Relations need to support:

- **ALL** – every defined item is required
- **ANY** – one item from a group is sufficient

- **ALL + ANY** – mandatory items and alternative groups can exist together

StockScope should calculate how many complete sets are currently available and show where their components are located.

The components do not need to be stored in the same location.

Quantity-Based Set Calculation

The warehouse data does not always tell us which specific tool is physically paired with which specific box.

For example:

10 boxes + 5 compatible tools = 5 possible complete sets

StockScope therefore needs to support calculations based on quantities, not only direct object-to-object relationships.

Find New Sets

StockScope should compare:

Empty Boxes + Loose Tools + Set Definitions → Possible New Sets

The user should be able to see:

- which sets can be created
- how many can be created
- required components
- available quantities
- locations of those components

Advanced Stock Control

Phase 2 should extend the basic Stock Control functionality with information calculated from relationships between warehouse items.

It should add:

- possible new sets
- incomplete sets
- missing set components
- other unusual conditions discovered through set and relationship analysis

The idea remains simple:

Instead of asking:

> **"What should I check?"**

StockScope should help answer:

> **"These are the things worth checking."**

Basic Historical Comparison

Because StockScope has already started preserving warehouse snapshots, Phase 2 can begin using this data for simple comparisons.

This could include:

- total stock growth or decline
- changes in object quantities
- changes in location occupancy
- new or removed objects
- basic location changes
- basic status changes

The purpose at this stage is not to build advanced analytics.

It is to start using the historical data that StockScope has already collected.

Phase 3 – Advanced Capacity, History & AI

Computed Capacity

Once reliable location and object dimensions are available, manual capacity can be replaced or supplemented by calculated capacity.

The calculation can use:

- location width / length / height
- box width / length / height
- quantity

StockScope can then calculate physical occupancy and available space more accurately.

Until then, manual capacity from Phase 1 remains the fallback.

Advanced Historical Analysis

Because warehouse snapshots are collected from the earlier phases, StockScope can gradually build a history of how the warehouse changes over time.

Later versions can use this data to analyse:

- new or removed objects
- location changes
- status changes
- stock growth or decline
- occupancy changes
- frequently moved objects
- locations that are repeatedly emptied and filled
- boxes that have not moved for a long time

Some simple comparisons, such as total stock growth or decline, may already become available earlier.

Phase 3 should focus on deeper historical analysis, trends and patterns rather than beginning the collection of historical data.

Scan History and Activity Analysis

Future versions of StockScope should support importing Scan Report data from the existing warehouse system.

Scan Report data can provide individual scan events, including:

- Item H-code
- scan date and time
- scan user
- recorded location
- recorded status
- Article
- department
- related Article information, when available

This data should be stored separately from warehouse snapshots and linked to Items using their H-codes.

Future functionality may include:

- individual Item Scan History
- chronological scan timelines
- recorded location and status changes
- scan activity statistics by user and time period
- automatic detection of suspicious scan patterns
- identification of potential workflow or data inconsistencies

Scan activity statistics should provide operational insight rather than automatically measuring individual employee performance.

Scan Reports may not represent every physical movement or guarantee that every recorded scan completed successfully.

Phase 1 does not require Scan Report integration.

Scan History and activity analysis are future enhancements that should be considered when designing the StockScope data models and persistence architecture.

AI Suggester

AI should be added only after StockScope has reliable data and analytics.

The purpose is not simply to add a chatbot.

AI should use information already calculated by StockScope and provide useful suggestions such as:

> **"6 additional complete sets can currently be created."**
>
> **"BS-F08 is approaching its capacity."**

"These boxes have not moved for a long period."

"These Push Items are currently available."

The normal StockScope business logic should remain deterministic.

AI should sit on top of that information and help identify useful patterns and possible actions.

Future Personal Project – Scanner UI Experiment

HQ Pack already has a company-wide Scanner integrated with its warehouse systems.

I do not want to replace or compete with it.

This would be a separate personal project where I could experiment with a modern scan-first warehouse workflow based on my own warehouse experience.

The default screen should always be ready for scanning while functions such as Stock Control, Batch Scanning or Search remain easily accessible.

After scanning an object, the scanner should immediately show:

- name / article
- status
- current location
- quantity in stock
- suggested location, when available
- note or warning
- available actions
- More Details

The basic principle would be:

SCAN → SEE INFORMATION → ACT

For a common location change:

Scan Object → Show Info + Suggested Location → Scan Actual Location → Confirmation → Enter → Ready for Next Scan

The suggested location is only guidance.

The worker still physically scans the actual destination location.

Other actions such as Change Status should be available immediately after scanning the object without requiring unnecessary navigation.

The interface should always clearly confirm what changed and then return directly to the ready-to-scan state.

As a small additional experiment, the time between the object scan and destination scan could also be used to estimate warehouse movement times.

This would remain a personal learning and portfolio project focused mainly on fast workflow and modern warehouse UI/UX.

Expected Improvements

StockScope should not only make warehouse information easier to see.

Where possible, its impact should also be measurable.

Some expected improvements include:

- less time spent searching through reports
- faster identification of stock requiring attention
- faster access to location and object information
- easier identification of available Push Items
- faster understanding of warehouse occupancy
- less manual comparison of warehouse data
- faster identification of possible complete sets in later phases

Some of these improvements can be partially quantified.

For example, the time required to find specific information using the current workflow can be estimated and later compared with the same task performed using StockScope.

Historical warehouse reports can also be used as test data to compare how easily information can be extracted using the old workflow and StockScope.

Possible measurements could include:

- time required to find a specific object
- time required to inspect a location
- time required to identify an unexpected status
- time required to find available Push Items
- time required to determine warehouse occupancy
- time required to identify components for a complete set

The purpose is not to prove that every improvement can be reduced to a number.

The goal is to have a practical way to evaluate whether StockScope actually makes warehouse work faster and easier.

Architecture Process

The development of StockScope follows eleven architectural steps:

- **Business Analysis**
- **User Flow**
- **Data Models**
- **Services & State**

- **Components**
- **Routing**
- **Implementation**
- **Validation**
- **Testing**
- **Data Flow Check**
- **Deployment & Production**

The three phases describe **what StockScope will gradually become**.

The eleven architecture steps describe **how it will be designed and built**.

1. Business Analysis

Why I Started StockScope

I work with warehouse stock every day and one problem I keep running into is that we have a lot of data, but getting a simple answer from that data can take too much time.

The information is there.

We know what boxes we have, where they are located and what they contain.

But when we need to make a decision, we often have to search through the system, filter data or export it to Excel and put the information together ourselves.

StockScope started as an idea to make this information easier to see and, more importantly, easier to use.

Problems I Want to Solve

How many complete sets do we have?

Knowing how many individual boxes we have is not always enough.

A product or shipment can require several different boxes or tools to create one complete set.

StockScope should automatically calculate how many complete sets can currently be created.

Can we build a set from stock spread across different locations?

The parts needed for a set do not necessarily have to be stored together.

StockScope should answer:

> **Do we currently have everything needed to make the set, and where can I find it?**

Where are our Push Items?

Push Items should be easy to find without searching through a large amount of stock data.

StockScope should show:

- what is available
- how much is available
- where it is located
- relevant status
- expected or planned sending day, when applicable
- useful operational notes

What is stored in each location?

A warehouse location can contain multiple objects and multiple article types.

Selecting a location should immediately show its contents, quantities, statuses and other relevant stock information.

How full is the warehouse?

StockScope should visualize location occupancy and available capacity.

Initially this can use manually configured capacity.

Later it can use physical location and object dimensions.

What requires attention?

Instead of manually checking every location and every stock record, StockScope should help identify exceptions.

This can include:

- unexpected statuses
- locations requiring attention
- available Push Items
- other unusual stock conditions

Later versions can add more intelligent checks based on relationships between boxes, tools and complete sets.

The First Goal

The first version of StockScope should answer simple practical questions:

- **What do we have?**
- **Where is it?**
- **What is inside this location?**
- **How full is this location?**
- **Is something here requiring attention?**
- **Which Push Items are available?**

- **How is overall warehouse occupancy changing over time?**
- **Is the total number of Items increasing or decreasing?**
- **How is the number of active Reviews changing over time?**

Later versions should also answer:

- **How many complete sets do we have?**
- **Can we create additional sets?**
- **Where are the required components?**
- **Which Items are frequently moved?**
- **Which locations have high turnover?**
- **Which Items have remained inactive for a long time?**

Phase 1 introduces basic historical trends using snapshots created after successfully confirmed Location Report imports.

More advanced historical analysis, including individual Item movements and long-term warehouse patterns, belongs to later phases.

The first step is simple:

> **Take the warehouse data we already have and turn it into information we can actually use.**

2. User Flow

The User Flow is currently focused on Phase 1.

The goal is to define how StockScope will be used during daily Stock Control work while keeping the application easy to extend in later phases.

Main Flow

The Phase 1 flow starts with the Location Report.

Location Report → Import → Validation → Preview → Compare → Confirm Import → Update Warehouse State → Reconcile Reviews → Create Snapshot → Dashboard

The imported report is first validated and compared with the current warehouse state before it can replace the active data.

A new warehouse snapshot is created only after the import has been successfully confirmed.

This prevents invalid or suspicious report data from automatically becoming part of the warehouse history.

StockScope keeps its own application data, such as review states, review history and user actions, separately from the imported report.

Dashboard

The Dashboard provides a quick overview of the current warehouse state.

The Phase 1 Dashboard includes:

- current warehouse occupancy
- number of Push Items
- number of Locations to Review
- latest successful import date and time
- Locations to Review overview
- occupancy trend
- Push Items overview
- quick access to import a new Location Report

The Dashboard always represents the warehouse state created by the latest successfully confirmed import.

If a new report is uploaded but fails validation or the import is cancelled before confirmation, the existing Dashboard data remains unchanged.

The Dashboard should show only the most useful information and provide navigation to more detailed views.

Some dashboard space can remain reserved for functionality introduced in later phases.

Locations to Review

StockScope identifies locations that may require attention.

The Dashboard shows approximately the top 8 Locations to Review with a `View All` option.

A location can have multiple review reasons at the same time.

Phase 1 review reasons:

Location-level

- **Low Occupancy** – occupancy ≤ 15%
- **Nearly Full** – occupancy ≥ 90% and < 100%
- **Full** – occupancy = 100%
- **Over Capacity** – occupancy > 100%

Capacity-related Review Reasons are mutually exclusive. A location should not receive multiple capacity Reviews for the same occupancy value.

Item-level

- **Wrong Status** – an Item has a status or status-location combination that violates a defined warehouse business rule.

- **Unexpected / Mixed Item** – an Item has an exceptional warehouse condition requiring Stock Control investigation, or does not match the expected contents or rules of its location.
- **Push Item** – an Item belonging to a configured Push Article that requires further action according to its Push workflow.

The exact occupancy thresholds are business rules and can be adjusted later.

Each review reason should have its own visual indicator, such as a small colored dot, so different reasons can be recognized quickly.

The `Locations to Review` count represents the number of unique locations with at least one active review, not the total number of individual reviews.

A single location can contain multiple active reviews while still counting as one Location to Review.

For example, if one location has an Over Capacity review and two item-level reviews, it still counts as one Location to Review while contributing three individual reviews to the total review count.

Location Reviews vs. Item Reviews

Location Reviews describe conditions affecting the location as a whole.

Examples include:

- Low Occupancy
- Nearly Full
- Full
- Over Capacity

Item Reviews describe conditions affecting individual H-code Items.

Examples include:

- Wrong Status
- Unexpected / Mixed Item
- Push Item

An Item Review does not automatically create a separate Location Review.

However, a location containing one or more Items with active Reviews should still appear in Locations to Review.

Location Detail displays Location Reviews separately from Item Reviews.

The total active Review count for a location includes both types, without duplicating individual Review records.

This allows StockScope to identify locations requiring attention while preserving the distinction between location-level and item-level problems.

Locations

The Locations page is the central place for browsing and reviewing warehouse locations.

StockScope uses one Locations page with different filter states instead of separate pages for different workflows.

The current filter and sorting state should be represented by query parameters.

Examples:

```
/locations
```

```
/locations?status=review&sort=priority
```

```
/locations?status=handled
```

This allows the same page to be opened from different parts of the application with the appropriate view already selected.

The page includes:

- reactive search by location code
- All, To Review and Handled views
- Review Reason filter
- minimum and maximum occupancy filter
- sorting
- location count

Review Reason filters should display the current number of matching locations.

Reasons with no current matches should remain visible but disabled.

Each location row should show:

- location code
- occupancy
- item count
- active reviews
- last checked

Hovering over the item count can provide a small preview of article names without opening the location.

Clicking a location opens the Location Detail page.

`Last Checked` represents the last time the location was physically inspected during Stock Control work.

A location can be marked as Checked independently from its review state.

For priority sorting, a location is ordered according to its highest-priority active review.

Current review priority from lowest to highest:

- Low Occupancy
- Nearly Full
- Unexpected / Mixed Item
- Wrong Status
- Push Item

- Full
- Over Capacity

`View All` from the Dashboard's Locations to Review section opens the same Locations page with the To Review filter and priority sorting already applied.

Future versions may expand reactive search to find locations by article or barcode, for example `NEWAYS56`, `H343791` or `DUONXE-03`.

Future versions should also evaluate whether users should be able to manually add a location to review when an issue is discovered during physical Stock Control work.

Location Detail

A Location Detail should display:

- location code
- current occupancy
- total number of Items
- active location-level reviews
- individual Items identified by their H-code
- Article associated with each Item
- current Item status
- active item-level reviews
- last physical check
- relevant location history

Location-level reviews should be displayed near the location overview, while item-level reviews should be displayed next to the affected Items.

Each Item can be opened in the Item Detail modal without leaving the Location Detail context.

Location Overview

The top of the page provides a compact overview of the location.

It should include:

- location code
- current occupancy
- total number of items
- number of active reviews
- articles currently present at the location
- last physical check
- capacity, when available

The active review count includes both location-level and item-level active reviews.

The article overview can show the article name together with the number of items, for example:

`FRENCKEN07 (27) · NEWAYS56 (15)`

The header should remain compact so that the item list is visible without unnecessary scrolling.

Location Reviews

Location-level reviews are displayed directly below the location overview.

Examples include:

- Low Occupancy
- Nearly Full
- Full
- Over Capacity

Each review can be handled individually using `Mark as Handled`.

If the location has no active location-level reviews, this section is not displayed.

Item-level reviews are not repeated in this section. They are displayed directly next to the affected item in the item list.

Items

The Location Detail page displays all individual items currently stored at the location.

The list is not limited to a preview. If a location contains 8, 22 or 46 items, all items should be available directly on the Location Detail page.

Each item row should include:

- item ID / H-code
- article
- status
- active review, when applicable
- `Handle` action for an active item-level review

For example:

| ID | Article | Status | Review |

| ------- | ---------- | -------- | ------------ |

| H145784 | FRENCKEN07 | Storage | — |

| H145785 | FRENCKEN07 | Storage | Push Item |

| H145786 | NEWAYS56 | Cleaning | Wrong Status |

Item-level reviews such as `Wrong Status`, `Unexpected / Mixed Item` and `Push Item` are therefore shown directly in the context of the affected item.

Item Filters

The item list can be filtered without leaving the Location Detail page.

Phase 1 should support:

- item ID search
- article filter
- status filter
- `All / Reviews only`

Articles available at the current location can be shown as selectable checkboxes with their item counts.

For example:

`☑ FRENCKEN07 (12)`

`☑ NEWAYS56 (7)`

`☑ DUONXE-03 (3)`

All articles are selected by default.

The user can select or deselect individual articles to quickly narrow the item list.

The page should show the number of currently displayed items, for example:

`Showing 7 of 22 items`

Filters should work together and update the visible item list without leaving the page.

Items Page Integration

Location Detail provides the item functionality needed for working with one specific location.

A separate Items page will provide a broader view for searching and filtering items across the warehouse.

The user can open the current Location Detail item results in the Items page while preserving the relevant filter context.

For example:

`/items?location=BS-A04`

or:

`/items?location=BS-A04&article=NEWAYS56`

This allows the user to move from a location-focused workflow into a broader item-focused workflow without losing context.

Related Articles and Sets

Phase 1 does not calculate complete sets.

However, the Location Detail design should remain compatible with future relations between articles.

For example, if `WETZLAR901` represents a box and `WETZLAR902` represents a related tool, both articles are displayed normally in Phase 1.

A later phase can use their relationship to provide information such as:

`5 complete sets · 15 empty boxes`

The future Items page should also be able to display multiple related articles together through filters.

Mark as Checked

`Mark as Checked` represents a physical Stock Control inspection of the location.

When the user marks a location as checked:

- the `Last Checked` timestamp is updated
- the physical check is recorded in location history

Marking a location as checked does not automatically handle or resolve its active reviews.

`Checked` therefore represents a physical inspection of the location, while `Handled` represents an action taken on a specific review.

Location History

The Location Detail page provides a compact history section.

Phase 1 should include a simple occupancy trend based on stored snapshots, for example:

`76% → 82% → 94% → 103%`

The page can also show recent activity such as:

- Over Capacity detected
- Location physically checked
- Wrong Status resolved

The goal is to provide useful recent context without turning the Location Detail page into a full analytics page.

More advanced historical analysis belongs to the dedicated Trends functionality and later development phases.

Navigation

When a user opens Location Detail from a filtered Locations view, returning to Locations should preserve the previous context whenever possible.

For example:

`/locations?status=review&sort=priority`

→ `BS-A04`

→ Back to Locations

The user should return to the same filtered and sorted Locations view instead of being returned to the default `All Locations` view.

This allows Stock Control work to continue without repeatedly rebuilding the same filters.

Item Detail

Selecting an individual H-code opens the Item Detail in a modal.

A separate Item Detail page is not required for Phase 1.

The modal allows the user to inspect a specific warehouse object without leaving the current page or losing the current filters, sorting or scroll position.

The same Item Detail modal can be opened from:

- Location Detail
- Items Page

Closing the modal returns the user to the same context from which it was opened.

Item Information

Phase 1 should include:

- H-code
- article
- customer
- customer article number
- current location
- location level
- current status
- dimensions, when available
- Call Off information, when applicable
- active item-level reviews
- article relations

The Location Report already provides additional item information such as customer article number, length, width, unit and kinds.

The `kinds` value does not need to be displayed directly.

When the imported `kinds` data identifies an item as belonging to a Call Off list, StockScope should extract and display the relevant Call Off information.

For example:

```
35 Call Off List ASML, RTM - Supplier Network
```

can be displayed as:

```
Call Off: ASML
```

If no Call Off information is present, the field should not be displayed.

When dimensions are available, they should be displayed in a compact form.

Example:

`1.60 × 1.20 m`

In Phase 1, dimensions are informational.

Later phases can reuse the same data for more advanced capacity calculations.

Relations

Relations are an important part of the Item Detail.

When relation data is available, StockScope should show the related article or articles directly in the modal.

Existing HQ data contains relation-related information such as `articleSet`. The exact relation model and available source data should be investigated during implementation.

In Phase 1, Article Relations should be displayed only when reliable relation data is available from an existing data source.

The Location Report alone may not provide enough information to identify these relationships.

If relation data is unavailable, the Relations section should remain hidden rather than displaying incomplete or assumed relationships.

Advanced relation logic and complete set calculations belong to Phase 2.

Phase 2 should extend Article Relations with:

- ALL / ANY requirements
- AND / OR combinations
- multiple related articles
- complete set calculations
- missing set components
- set availability analysis

Item Notes

StockScope should allow users to add a simple note to an individual Item.

Notes provide operational context that is not available in the original warehouse report.

For example, an Item may have the status `On hold`.

After checking with a supervisor, the user learns that the Item is waiting for a response from customer ZEISS.

The user can add a note:

> Waiting for customer ZEISS response. Confirmed with supervisor.

The original Item status remains unchanged.

Note Management

Each Item should support one editable note in Phase 1.

The Item Detail Modal should allow the user to:

- view the current note
- add a note
- edit an existing note
- clear a note when it is no longer relevant

StockScope should store:

- Item H-code
- note text
- last modified date and time

The note belongs to the individual H-code Item, not to the Article.

Data Persistence

Item Notes are StockScope-owned data.

They must remain available after importing a new Location Report.

An Excel import must not overwrite or delete existing Item Notes.

If an Item temporarily disappears from the Location Report, its note should remain stored.

If the same H-code appears again in a later report, the existing note should still be available.

Relationship to Reviews

Item Notes are independent of the Review Workflow.

Adding, editing or clearing a note must not:

- change the original warehouse status
- create a Review
- mark a Review as Handled
- resolve an existing Review

An Item may have an Attention indicator and an Item Note without having an active Review.

Phase 1 Scope

Phase 1 should keep Item Notes simple.

Advanced features such as multiple notes, comments, case management, attachments and note history are not required.

Item Reviews

Item-level reviews should be displayed directly in the Item Detail modal.

Examples include:

- Wrong Status
- Unexpected / Mixed Item
- Push Item

When a review has been addressed physically, the user can use:

```
Mark as Handled
```

Handled does not mean that StockScope has verified that the problem is resolved.

The review remains waiting for verification until a new warehouse report confirms whether the issue has disappeared or is still present.

Article Image

An Article may optionally have an image that can be displayed when viewing an Item.

Because multiple physical Items can belong to the same Article, the image should be associated with the Article rather than with an individual H-code.

Article images are optional and are not required for Phase 1.

If a reliable image source becomes available later, the same Article image can be reused across all Items belonging to that Article.

Future Item Data Integration

Future versions of StockScope should evaluate additional item information available from the existing HQ system.

Potential integrations include:

- Scan History
- additional article relation data
- existing article images
- associated documents such as work instructions

Future Scan History should provide item scan history where available, including:

- scan user
- previous scans
- status changes
- location movement

This could help answer who scanned an item, when it was scanned, where it was previously located and what its previous status was.

These additional integrations are not required for Phase 1.

Phase 1 should remain functional using the Location Report as its primary data source.

Item Status Classification

StockScope uses the original Item statuses from the existing warehouse management system.

The application must not modify the original status values.

Status Classification determines how an Item is interpreted and displayed.

Review Detection determines whether the Item requires Stock Control attention.

An Item can have a valid location and still require a Review because of its status.

Normal Storage Statuses

The following statuses represent healthy warehouse storage conditions:

- `Storage`
- `Storage - Cleanroom clean`
- `RTM- New Build`
- `Storage - Reserved for ASML Factory`

These statuses do not automatically create Reviews.

The existing visual distinctions should remain:

- `Storage` – Blue, Grade 5 storage
- `Storage - Cleanroom clean` – Pink, Grade 4 storage
- `RTM- New Build` – Red, special healthy storage version
- `Storage - Reserved for ASML Factory` – Valid reserved storage

The red color used for `RTM- New Build` must not indicate an error.

Damage and Repair Workflow

The following statuses are valid only when the Item is assigned to the location `Transfer to Run 2`:

- `Ready for Repair`
- `Ready for Inspection at Tech Repair`

`Ready for Repair` is used for damaged empty boxes.

`Ready for Inspection at Tech Repair` is used for damaged tools or boxes containing damaged tools.

Examples:

- `Ready for Repair` + `Transfer to Run 2` → Valid
- `Ready for Inspection at Tech Repair` + `Transfer to Run 2` → Valid
- `Ready for Repair` + `BS-A03` → Wrong Status Review
- `Ready for Inspection at Tech Repair` + `BS-A03` → Wrong Status Review

The Review Engine must validate the combination of status and location.

These statuses must not automatically create Reviews when the Item is in the correct transfer location.

Transport Statuses

The following statuses represent transport-related workflow states:

- `Transported to ASML`
- `Transported to customer`

Their expected location is:

`EXTERN-NL-EIN-01`

A transport status combined with a normal warehouse storage location is invalid.

For example:

- Status: `Transported to ASML`
- Location: `BS-A03`
- Result: Wrong Status Review

The same rule applies to `Transported to customer`.

This can happen when an Item receives its transport status but its location is not updated.

Items correctly scanned to `EXTERN-NL-EIN-01` are no longer included in the next Location Report.

Therefore, Phase 1 cannot directly verify their final transport state.

Wrong Status Reviews

The following statuses require a Wrong Status Review when an Item remains present in the monitored warehouse:

- `Cleanroom Cleaning`
- `Industrial Cleaning`
- `Ready for Cleanroom Cleaning`
- `Ready for Industrial Cleaning`
- `Ready for Storage`
- `Ready for Transport`
- `Industrially cleaned`
- `Stock Correction`

Items with these statuses require a Wrong Status Review and should allow the user to add an explanatory Item Note.

Adding a note does not automatically resolve the Review.

The Review remains active until a later confirmed import verifies that the condition no longer exists.

These statuses indicate that the Item is not in an expected final storage state.

A Wrong Status Review should allow Stock Control to investigate why the Item remains in the warehouse with that status.

Transport statuses also create Wrong Status Reviews when combined with an incorrect location.

The Review Engine must preserve the original status and record the detected Review Reason separately.

Unexpected / Mixed Item Reviews

The following statuses require an Unexpected / Mixed Item Review:

- `Inspected - Waiting for Tech Repair Parts`
- `Intern repair CE`

- `On hold`
- `Pending PE`
- `Waiting for Parts`
- `Tech Repairing`
- `Scrapped`

These statuses may represent legitimate operational situations.

However, their presence in the monitored warehouse requires investigation or explanation.

A valid warehouse location does not automatically remove the need for a Review.

For example:

- Status: `On hold`
- Location: `BS-A03`
- Result: Unexpected / Mixed Item Review

The user can investigate the reason and add an Item Note.

Example:

> Waiting for customer response. Confirmed with supervisor.

Adding a note does not automatically resolve the Review.

The Review remains subject to the normal Review Workflow and subsequent warehouse data verification.

Attention Indicators

Attention indicators provide additional visual information about Items requiring investigation.

An Item with an Unexpected / Mixed Item Review may also display an orange Attention indicator.

The Attention indicator is not a separate Review Reason.

StockScope must avoid creating duplicate Reviews for the same detected condition.

Review Detection Principles

Review Detection must follow defined warehouse business rules.

The Review Engine should consider:

- original Item status
- current Item location
- warehouse workflow context
- configured Review rules

The following distinctions are important:

- Normal Storage Status: healthy storage condition
- Damage Workflow Status: may be valid in the correct repair context
- Transport Status: valid only with the expected external location

- Wrong Status: a defined invalid or unfinished warehouse status condition
- Unexpected / Mixed Item: an exceptional condition requiring investigation

A Review must not change the original warehouse status.

Reviews, Attention indicators and Item Notes remain separate StockScope concepts.

Status Display

Status classification and Review indicators should be displayed consistently in:

- Items Page
- Location Detail
- Item Detail Modal

The original status text must remain visible.

Colors should not be the only way to communicate meaning.

Phase 1 Scope

Phase 1 should support:

- recognition of known warehouse statuses
- visual distinction between healthy storage statuses
- automatic Wrong Status Review detection
- automatic Unexpected / Mixed Item Review detection
- transport status and location validation
- Item Notes for operational explanations
- integration with the existing Review Workflow

Advanced scan verification remains outside Phase 1.

Review Workflow

Reviews represent specific problems detected by StockScope.

A review is not automatically considered resolved when a user performs an action.

The basic review lifecycle is:

OPEN → Mark as Handled → WAITING FOR VERIFICATION → RESOLVED

A user can mark an individual review as `Handled`.

This means that the user has taken action on the specific problem, but StockScope still needs to verify the result using new warehouse data.

The action is stored by StockScope and is not overwritten by a new Excel import.

After the next successfully confirmed import, StockScope evaluates the review condition again.

If the problem no longer exists:

WAITING FOR VERIFICATION → RESOLVED

If the problem is still detected:

WAITING FOR VERIFICATION → OPEN

A review can also become resolved without being manually handled.

Normal warehouse activity can remove the condition that originally created the review.

In this case:

OPEN → RESOLVED

For example, a Low Occupancy review may disappear because additional items were moved into the location during normal warehouse activity.

A Push Item review may disappear because the item was moved out of the warehouse before anyone manually marked the review as Handled.

In these situations, the review history should record that the problem was resolved by a warehouse data change rather than by a user action.

Reviews belong to the specific problem, not automatically to the entire location.

This allows one location to contain several independent reviews with different states.

Push Item Verification

Push Item Reviews should be evaluated according to the workflow type configured for each Push Article.

Phase 1 supports two Push workflows:

- ASML Push
- Outbound Push

Both workflows use the same external destination location, but they have different final statuses and sending schedules.

ASML Push Items

The expected final warehouse state is:

- Location: `EXTERN-NL-EIN-01`
- Status: `Transported to ASML`
- Expected sending day: Every day

ASML Push Items may require action on any working day.

Outbound Push Items

The expected final warehouse state is:

- Location: `EXTERN-NL-EIN-01`
- Status: `Transported to customer`
- Expected sending day: Configurable day of the week

The expected sending day is maintained as part of the Push Article configuration.

StockScope can visually highlight an active Outbound Push Item when its expected sending day is approaching or has arrived.

The sending day affects operational priority but does not create a separate Review Reason.

Location Report Visibility

Items transferred to `EXTERN-NL-EIN-01` are no longer included in the available Location Report.

Therefore, StockScope cannot directly verify the final external location or transport status using the Location Report alone.

The absence of an Item from the report does not prove that the Item was transported to ASML or to a customer.

StockScope must distinguish between:

- an Item no longer present in the monitored warehouse
- an Item whose final transport has been independently verified

Push Item Review Resolution

The Location Report always represents the complete Best warehouse.

Items scanned to `EXTERN-NL-EIN-01` are no longer included in the Location Report.

After a successfully validated and confirmed import, StockScope compares the new warehouse state with the previous confirmed state.

If an Item with an active Push Item Review is no longer present in the new report, its Review automatically becomes RESOLVED.

The resolution reason is:

```
No longer present in monitored warehouse
```

A manual Mark as Handled action is not required.

This does not verify that the Item was scanned to `EXTERN-NL-EIN-01` or transported to ASML or a customer.

Suspicious import changes must be presented to the user before confirmation. Structurally invalid reports must be rejected.

Review History must remain preserved.

Returning Items

An Item may later return to the monitored warehouse and appear in a new Location Report.

If the returning H-code still matches an active Push Article configuration and requires attention, StockScope should create a new Push Item Review.

The previous resolved Review and its history must remain preserved.

Review Lifecycle

Mark as Handled records that a user has addressed the Push Item Review.

It does not confirm that the Item has left the warehouse.

The Review remains `WAITING FOR VERIFICATION` until a newer confirmed report provides sufficient evidence to evaluate the condition.

If the Item is still present and continues to require action, the Review returns to `OPEN`.

If the Item is no longer present in the next successfully confirmed Location Report, its Review automatically becomes RESOLVED.

Resolution reason:

```
No longer present in monitored warehouse
```

A Push Item Review may also become resolved without a manual Handled action when the required warehouse data change is detected.

Future Verified Transport

A future reliable data source may provide evidence of the actual final location and status.

For example, a future Scan Report integration may provide additional scan information.

When both the destination and final status can be reliably verified, StockScope may distinguish verified transport completion from disappearance from the monitored warehouse.

Scan Report integration and verified transport history are not required for Phase 1.

More advanced Push Rules based on Article Relations and complete sets remain part of Phase 2.

Wrong Status Context

Wrong Status and Unexpected / Mixed Item Reviews follow the rules defined in Item Status Classification.

Review Detection must consider the original Item status, location and applicable warehouse business rules.

Normal storage statuses do not automatically require Reviews.

`Ready for Repair` and `Ready for Inspection at Tech Repair` are valid only at `Transfer to Run 2`. At other locations, they create a Wrong Status Review.

Transport statuses require special validation.

For example:

```
Transported to ASML + BS-A03 → Wrong Status Review
```

```
Transported to customer + BS-A03 → Wrong Status Review
```

Both statuses expect the external location:

```
EXTERN-NL-EIN-01
```

Unexpected / Mixed Item statuses such as `On hold`, `Pending PE` and `Waiting for Parts` require investigation even when the current warehouse location is valid.

An Item Note can provide an explanation without changing the original status or automatically resolving the Review.

The Review Engine must not generate duplicate Reviews for the same detected condition.

Unknown or unclassified statuses should remain visible until reliable business rules are defined.

Items

Items will have their own main page accessible from the sidebar.

The Items page provides a global view of individual warehouse objects across all locations.

Each row represents one specific H-code.

The page should make it possible to find individual objects and compare items across multiple articles, locations, statuses and review states without navigating through individual locations first.

Search

The Items page should provide a single reactive search field.

Phase 1 search should support:

- H-code
- article
- customer article number

Examples:

`H398913`

`ZEISS022`

`4022.674.1152x`

Filters

The Items page should support multi-select filtering.

Phase 1 filters include:

- Article
- Location
- Status
- Review
- Size
- Call Off

Multiple values can be selected within the same filter.

Values inside one filter group use OR logic.

Different filter groups are combined using AND logic.

Example:

```
(ZEISS021 OR ZEISS022) AND (BS-A01 OR BS-A04) AND Storage AND Large
```

Active filters should remain visible and can be removed individually.

A `Clear All` action resets all active filters.

The page should also show how many items match the current filters.

Example:

```
Showing 37 of 4,382 items
```

Size Filter

Size is an Article property.

The Location Report already provides:

- length
- width
- unit

Phase 1 should provide simple size filters:

- Small
- Medium
- Large

The exact boundaries between these categories should not be defined until the real article dimension data has been analyzed.

This avoids creating arbitrary size rules before understanding the actual warehouse data.

Call Off

Call Off is an Article property rather than a property of an individual H-code.

When an Article belongs to a Call Off list, a small visual indicator should be displayed next to the Article in the Items table.

The indicator can provide additional information on hover.

Example:

```
Call Off: ASML
```

Call Off should also be available as a filter.

Selecting a Call Off filter shows all H-codes whose Article belongs to the selected Call Off.

A separate Call Off table column is not required.

Items Table

Each table row represents one individual warehouse object.

Phase 1 should show:

- H-code
- article
- customer article number
- location
- status
- active review indicator

Selecting the H-code or item row opens the Item Detail modal.

Closing the modal returns the user to the same Items page context without losing filters, sorting or scroll position.

Location Preview

The Location value in an item row should provide a quick preview on hover.

The preview can include:

- occupancy
- number of items
- number of active reviews
- last checked
- article counts

The preview is informational only and should not contain editing or review actions.

Selecting the Location opens its Location Detail page.

Query Parameters

The Items page should use query parameters to represent filters and sorting where appropriate.

Examples:

```
/items
```

```
/items?location=BS-A04
```

```
/items?location=BS-A04&article=NEWAYS56
```

```
/items?status=storage
```

This allows other parts of StockScope to open the same Items page with useful filters already applied.

For example, `Open in Items Page` from Location Detail can open:

```
/items?location=BS-A04
```

The same Items page is therefore reused instead of creating separate pages for different item views.

Article and Item Data

StockScope should distinguish between information belonging to an Article and information belonging to an individual warehouse object.

Article-level information can include:

- customer article number
- size
- Call Off
- relations
- image, when available

Individual H-code information can include:

- location
- level
- status
- active reviews

This distinction should be reflected later in the StockScope data models.

Push Items

Push Items will not have a separate page in Phase 1.

The existing Items page will be reused with a Push Items quick filter.

This keeps the workflow simple and avoids duplicating item search, filtering and table functionality.

Push Item Definition

In Phase 1, Push Items are identified using StockScope-owned Push Article configuration.

Each Push Article defines:

- Article
- Push workflow type (ASML or Outbound)
- expected sending day
- optional note

StockScope identifies individual H-code Items belonging to configured Push Articles and evaluates their current warehouse state against the corresponding workflow rules.

Phase 1 supports two Push workflows:

ASML Push

- Expected sending day: Every day
- Expected final status: `Transported to ASML`
- Expected final location: `EXTERN-NL-EIN-01`

Outbound Push

- Expected sending day: Configurable day of the week
- Expected final status: `Transported to customer`
- Expected final location: `EXTERN-NL-EIN-01`

An Item receives an active Push Item Review only when it requires further action according to its configured Push workflow.

Items transferred to the external location are no longer included in the available Location Report.

Therefore, StockScope cannot directly verify their final transport status or destination using the Location Report alone.

When an Item with an active Push Item Review disappears from the next successfully confirmed Location Report, its Review automatically becomes RESOLVED.

Resolution reason:

```
No longer present in monitored warehouse
```

Because the Location Report represents the complete Best warehouse, the Item is considered no longer present in the monitored warehouse.

This does not verify its final external location or transport status.

Suspicious changes must be reviewed during Import Validation before the new report is confirmed.

Push Article configuration remains independent from imported Location Report data and must be preserved during future imports.

Future Push Rules can use Article Relations and Set Intelligence in Phase 2.

Push Items Filter

The Items page should provide a quick filter for active Push Item Reviews.

Selecting this filter shows individual H-code Items that currently require attention according to their configured Push workflow.

Items whose Push Item Reviews have already been resolved are not included in the active Push Items filter.

The Dashboard Push Items count and View All action should use the same active Push Item Review definition.

Example:

```
/items?review=push-item
```

The existing Items page filters remain available while the Push Items filter is active.

Push Articles

In Phase 1, Push Item configuration belongs to the Article rather than to an individual H-code.

If an Article is configured as a Push Article, all individual H-codes belonging to that Article can be identified as Push Items.

Example:

`ICHOR06 → Push Article`

All H-codes with Article `ICHOR06` can therefore appear in the Push Items view.

This avoids manually configuring individual warehouse objects.

Manage Push Articles

Push Articles should be managed through a modal opened from the Items page.

The modal provides a simple editable list of configured Push Articles.

Each entry contains:

- Article
- Push workflow type (ASML or Outbound)
- Expected sending day
- Note (optional)
- Last changed date (automatic)

The expected sending day depends on the selected Push workflow type.

For ASML Push Articles, the sending schedule is automatically set to `Every day`.

For Outbound Push Articles, the user can select an expected day of the week.

If no weekday is configured for an Outbound Push Article, the Item can still receive a Push Item Review, but schedule-based indicators should not be displayed.

Users should be able to:

- add a Push Article
- edit an existing Push Article
- remove a Push Article

Changing the Push workflow type should update the available sending day options.

The Last Changed date should be updated automatically when a Push Article configuration is modified.

Push Article configuration must remain stored independently from imported warehouse data.

A separate Push Items management page is not required in Phase 1.

Expected Sending Day

Expected Sending Day defines when a Push Item is normally expected to be sent from the warehouse.

The sending schedule depends on the configured Push workflow type.

ASML Push

ASML Push Items use a daily sending schedule.

- Expected sending day: Every day
- No individual weekday configuration is required

Outbound Push

Outbound Push Items use a configurable day of the week.

Available options include:

- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday
- Sunday

The expected sending day can be configured individually for each Outbound Push Article.

If no sending day is configured, the Item should remain visible as an active Push Item when it requires action, but no schedule-based priority indicator should be displayed.

Sending Day Indicators

StockScope can visually highlight active Push Items based on their expected sending schedule.

For example:

- One day before the expected sending day: `Sending Soon`
- On the expected sending day: `Expected Today`

For ASML Push Items, the daily schedule does not require a separate `Sending Soon` indicator.

Sending Day indicators provide operational context and help users prioritize work.

They must not:

- create a separate Review Reason
- automatically mark a Review as Handled
- resolve a Push Item Review
- change the original warehouse status

The exact indicator styling can be decided during UI implementation.

Notes

Each Push Article can optionally contain a short operational note.

Examples could include destination, priority or other useful context.

The note should provide operational information without turning StockScope into a planning system.

Last Changed

StockScope should record when the Push Article configuration was last changed.

This makes it possible to understand how current the manually maintained Push Item information is.

StockScope Business Data

Push Article configuration is StockScope-managed business data.

It should be stored independently from imported warehouse reports.

The Location Report describes the current physical warehouse state, including which H-codes and Articles exist and where they are located.

StockScope separately maintains information such as:

- whether an Article is a Push Article
- Push workflow type (ASML or Outbound)
- expected sending day
- note
- last changed

Importing a new Location Report should therefore not remove the Push Article configuration.

Future Push Rules

Phase 1 uses a deliberately simple rule:

```
Article → Push Article
```

Future versions can extend this using Article Relations and Set Intelligence.

For example, a future Push Rule could require:

```
ICHOR06 AND ICHOR07 → Complete Set → Push
```

More advanced rules can later use the existing AND / OR relation logic.

This belongs to Phase 2 Warehouse Intelligence and should not complicate the Phase 1 Push Items workflow.

Review History

StockScope keeps review actions and results in its own application data.

This allows the application to remember that a review was previously handled even after another Location Report is imported.

A simple history can record events such as:

- review detected
- marked as handled
- verified as resolved
- detected again

Resolved reviews no longer need to appear in the active Locations to Review list but remain available in history.

Snapshots

Each successful Location Report import can create a warehouse snapshot.

A snapshot represents the warehouse state at a specific point in time.

Multiple snapshots can exist during the same day.

For example:

08:00 → Snapshot 12:00 → Snapshot 16:00 → Snapshot

All snapshots can remain stored.

For simple daily trends, StockScope uses:

- the latest available snapshot for the current day
- the final snapshot of previous days

This allows StockScope to preserve detailed import history while keeping the Phase 1 Trends view simple.

Snapshot Data

A snapshot should preserve enough information to support Phase 1 historical trends.

This includes:

- timestamp
- warehouse occupancy
- total item count
- total active review count
- active review counts by Review Reason

Review counts should include both Location Reviews and Item Reviews.

Location Reviews:

- Low Occupancy
- Nearly Full
- Full
- Over Capacity

Item Reviews:

- Wrong Status
- Unexpected / Mixed Item
- Push Item

Historical snapshot values should not be recalculated when the current warehouse state changes.

A snapshot represents what StockScope detected at that point in time.

For example:

Monday:

```
Push Items: 14
```

Tuesday:

```
Push Items: 9
```

Even if some of Monday's Push Item reviews are later resolved, the Monday snapshot should still preserve the historical value of 14.

Trends

The Trends page provides a simple historical overview of how the warehouse changes over time.

Phase 1 focuses on three main trends:

- Warehouse Occupancy
- Total Items
- Reviews

The goal is to provide useful operational history without turning Phase 1 into a complex analytics system.

Time Period

Users should be able to change the displayed period.

Initial options can include:

- 7 days
- 30 days
- 90 days

Warehouse Occupancy

Warehouse Occupancy shows how the overall warehouse occupancy changes over time.

The chart uses historical snapshot data.

For example:

```
61% → 64% → 63% → 67% → 68%
```

The page can also show:

- current value
- value at the beginning of the selected period
- change over the selected period

Total Items

Total Items shows how the total number of individual warehouse objects changes over time.

Each concrete H-code counts as one item.

For example:

`4,210 → 4,238 → 4,301 → 4,276 → 4,382`

This provides a simple indication of whether the physical stock in the warehouse is increasing or decreasing.

Reviews

The Reviews trend shows the number of active reviews recorded in each snapshot.

It includes both Location Reviews and Item Reviews.

Location Reviews include:

- Low Occupancy
- Nearly Full
- Full
- Over Capacity

Item Reviews include:

- Wrong Status
- Unexpected / Mixed Item
- Push Item

Users should be able to filter the Reviews trend by Review Reason.

Multiple Review Reasons can be selected when useful.

For example, selecting only `Push Item` shows the historical number of active Push Item reviews.

Selecting multiple reasons shows the combined trend for the selected Review Reasons.

Handled and Resolved Reviews

`Handled` and `Resolved` represent different events.

`Handled` means that a user explicitly indicated that they addressed a review.

`Resolved` means that the review condition no longer exists according to newer warehouse data.

A review does not need to be manually marked as Handled before it can become Resolved.

For example, a location can have Low Occupancy in one snapshot.

No user marks the review as Handled.

Later warehouse activity fills the location.

The next report no longer meets the Low Occupancy condition.

The review can therefore become:

`Resolved by warehouse data change`

The same principle applies to Item Reviews.

For example, if a Push Item is present in one report but is no longer in the warehouse in the next report, its review can become Resolved automatically.

The historical snapshot still preserves that the Push Item existed previously.

Review Verification

A review marked as Handled should remain waiting for verification until newer warehouse data is available.

The normal flow can be:

`OPEN → Mark as Handled → WAITING FOR VERIFICATION → RESOLVED`

If the condition is still present after the next import, the review can become active again.

Reviews can also follow:

`OPEN → RESOLVED`

when the condition disappears through normal warehouse activity without a manual Handled action.

This distinction allows StockScope to separate user actions from changes detected directly in warehouse data.

Dashboard Integration

A small trend overview can be displayed on the Dashboard.

Selecting `View Trends` opens the dedicated Trends page.

The full Trends page provides the longer historical view and filtering options.

Future Analytics

More advanced warehouse analytics are outside the Phase 1 Trends scope.

Future versions can evaluate metrics such as:

- frequently moved items
- location churn
- long-term inactive items
- article growth
- set trends
- capacity trends
- AI-assisted warehouse insights

Import Data

The Import Data page allows the user to import the latest Location Report into StockScope.

The import process should be simple, but it must also protect the application from incorrect or incomplete warehouse data.

A technically valid Excel file can still contain unrealistic or incorrect data.

For this reason, StockScope should not update the warehouse state immediately after a file is selected.

The Phase 1 import flow is:

`Select File → Validate → Process Preview → Compare → Confirm Import → Update Warehouse State → Reconcile Reviews → Create Snapshot → Import Complete`

Select File

The user can select or drag and drop a Location Report.

Phase 1 expects the existing Location Report in `.xlsx` format.

The page can also display information about the latest successful import, including:

- import date and time
- number of locations
- number of items

File Validation

StockScope should validate the file before allowing an import.

The initial validation should verify that:

- the file can be read
- the file is a supported Excel file
- the expected Location Report structure exists
- required columns are present

If required data or columns are missing, the import should be blocked.

For example:

`Missing required column: location`

In this situation, `Confirm Import` should not be available.

Import Preview

A valid file should first be processed as a preview.

Processing the preview must not modify the persistent StockScope state.

No snapshot should be created and no existing warehouse data or reviews should be changed at this stage.

The preview should show the most important values from the new report and compare them with the current warehouse state.

For example:

| Metric | Current | New Report | Change |

| --- | --- | --- | --- |

| Locations | 184 | 184 | 0 |

| Items | 4,345 | 4,382 | +37 |

| Occupancy | 66% | 68% | +2% |

| Reviews | 27 | 21 | -6 |

This gives the user an opportunity to verify that the new report looks reasonable before it becomes part of StockScope history.

Import Warnings

StockScope should detect unusually large changes when possible.

For example:

- extremely large decrease in item count
- occupancy unexpectedly dropping close to zero
- unusually large change in the number of locations
- other major differences compared with the current warehouse state

A warning means that the report is technically valid but the data change looks unusual.

For example:

```
Item count decreased by 99.5%.
```

```
Warehouse occupancy decreased from 66% to 0%.
```

The user should be asked to verify that the correct Location Report was selected.

Warnings should not automatically block the import because a large warehouse change can still be legitimate.

The user can still choose `Confirm Import`.

Errors and Warnings

Errors and warnings represent different situations.

An `Error` means that StockScope cannot reliably process the report.

The import must be blocked.

A `Warning` means that the report can be processed, but the resulting warehouse state looks unusual.

The user can still confirm the import after reviewing the preview.

Confirm Import

Only `Confirm Import` should allow the preview data to become part of the persistent StockScope state.

If the user cancels the preview:

- the current warehouse state remains unchanged
- no reviews are changed
- no snapshot is created

- no historical trend data is affected

This prevents an accidentally selected or incorrect report from contaminating warehouse history.

A confirmed import must be processed as one consistent operation.

Updating the warehouse state, reconciling Reviews and saving historical records must either complete successfully together or leave the previously confirmed state unchanged.

A partially completed import must not become the active warehouse state.

Repeated imports of the same report should be detected to prevent accidental duplicate historical records.

Update Warehouse State

After confirmation, the Location Report is processed through the StockScope data adapter and converted into the internal StockScope models.

The confirmed import updates the current warehouse state.

This includes warehouse-derived data such as:

- locations
- items
- statuses
- occupancy
- other values derived from the Location Report

The import should update warehouse data without deleting StockScope-specific application data.

The following StockScope data should remain persistent across imports:

- Push Article configuration
- Handled review actions
- Last Checked information
- Review History
- user notes
- historical snapshots

Review Reconciliation

After a successful confirmed import, StockScope evaluates Review conditions against the new warehouse data.

This can result in:

- new Reviews being detected
- existing Reviews remaining OPEN
- Handled Reviews returning to OPEN when their condition persists
- Reviews becoming RESOLVED when reliable new data confirms that their condition no longer exists

A Review does not need to be manually marked as Handled before it can become Resolved.

For example, a Low Occupancy Review can become RESOLVED when the next confirmed report shows that the location is no longer below the configured occupancy threshold.

The Location Report represents the complete Best warehouse.

After a successfully confirmed import, StockScope compares the new warehouse state with the previous confirmed state.

If an Item with an active Push Item Review is no longer present in the new report, the Review automatically becomes RESOLVED.

Resolution reason:

`No longer present in monitored warehouse`

This does not verify transport to ASML or to a customer.

Missing Items must be evaluated only after the import has passed validation and has been confirmed by the user.

Suspicious changes should generate import warnings before confirmation.

All Review transitions should be recorded in Review History.

Review reconciliation must preserve previous user actions, including Mark as Handled, and must not overwrite StockScope-owned notes or configuration.

Missing Item Verification

The Location Report represents the complete Best warehouse.

The report contains Items currently registered in the monitored warehouse.

Items scanned to the external location `EXTERN-NL-EIN-01` are no longer included in the Location Report.

After a new Location Report has been successfully validated and confirmed, StockScope compares the imported Items with the previous warehouse state.

If an Item with an active Push Item Review is no longer present in the new report, its Review automatically become RESOLVED.

For example:

Previous Report:

- H-code: H343791
- Article: ICHOR06
- Location: BS-A03
- Status: Storage
- Push Item Review: OPEN

New Report:

- H-code H343791 is no longer present

Result:

`OPEN → RESOLVED`

Resolution reason:

`No longer present in monitored warehouse`

This resolution does not require the user to manually mark the Review as Handled.

The Location Report does not directly confirm that the Item was scanned to `EXTERN-NL-EIN-01`.

Therefore, StockScope must not classify the resolution as verified transport.

The previous Review and its history must remain preserved.

If the same H-code later returns to the warehouse and matches an active Push Article configuration, StockScope can create a new Push Item Review.

Import Safety

StockScope should validate the imported report and warn users about suspicious changes.

Examples include:

- unusually large decreases in Item count
- unexpected changes in location coverage
- unrealistic occupancy changes
- missing or invalid required data

A structurally invalid report must be rejected.

A structurally valid report with suspicious changes should display a warning explaining the detected differences.

The user can then:

- Confirm Import
- Upload Another File

After successful confirmation, the imported report becomes the current warehouse state and normal Review Reconciliation is performed.

The application should not require individual confirmation for every missing Push Item.

Review Business Rules

Review detection must follow the warehouse business rules defined in Item Status Classification.

The Review Engine must evaluate:

- original Item status
- current Item location
- applicable warehouse workflow rules
- configured Push Article rules

Normal Storage statuses do not automatically create Reviews.

Damage and Repair statuses are valid only at `Transfer to Run 2`. When assigned to an incorrect location, they create a Wrong Status Review.

Transport statuses are valid only at `EXTERN-NL-EIN-01`. When they appear at a normal warehouse location, they create a Wrong Status Review.

Defined Wrong Status conditions create a Wrong Status Review.

Exceptional statuses such as `On hold`, `Pending PE`, `Waiting for Parts` and `Scrapped` create an Unexpected / Mixed Item Review, even when their location is otherwise valid.

An Item Note may explain the situation but must not automatically resolve the Review.

Review detection must preserve the original warehouse status.

Unknown statuses should remain visible without automatically being classified as healthy or incorrect.

Review rules should remain independent from the imported Excel structure.

Create Snapshot

A new warehouse snapshot should be created only after a successful confirmed import.

The snapshot preserves the warehouse state required by Phase 1 Trends.

This includes values such as:

- warehouse occupancy
- total item count
- total active review count
- active review counts by Review Reason
- snapshot timestamp

Cancelled previews and failed imports must not create snapshots.

Import Complete

After a successful import, StockScope should display a short summary.

For example:

```
184 Locations

4,382 Items

21 Reviews

68% Warehouse Occupancy
```

The summary can also show changes compared with the previous snapshot.

For example:

```
Items: +37

Reviews: -6
```

`Occupancy: +2%`

The user can then return to the Dashboard.

Phase 1 Scope

Phase 1 should focus on preventing incorrect data from entering StockScope rather than providing advanced tools for repairing warehouse history afterward.

The main protection is:

`Validate → Preview → Compare → Confirm`

Advanced snapshot management or manual historical data repair can be evaluated in later development phases.

Phase 1 Navigation

The initial main navigation is:

- Dashboard
- Locations
- Items
- Trends
- Import Data

Push Items do not require a separate main navigation entry in Phase 1.

They are accessed through the Items page using the Push Item filter.

For example, selecting `View All` from the Push Items section on the Dashboard opens the Items page with the Push Item filter already applied.

The navigation should remain simple and can be extended when functionality from later phases is introduced.

Phase 1 User Flow

The Phase 1 User Flow connects the main StockScope functions into one daily workflow.

The process starts with importing the latest Location Report.

Location Report → Import → Validation → Preview → Compare → Confirm Import → Update Warehouse State → Reconcile Reviews → Create Snapshot → Dashboard

After a successful import, the Dashboard becomes the main starting point for working with the current warehouse state.

From the Dashboard, the user can continue through three main areas:

Dashboard → Locations

The Locations page provides the warehouse overview and allows the user to find locations that require attention.

Locations → Location Detail → Item Detail Modal

Location Detail shows the current contents of the location, its occupancy, active reviews and history.

Individual H-code items can be inspected through the Item Detail modal without leaving the Location Detail context.

Dashboard → Items

The Items page provides access to individual H-code objects across the warehouse.

Items can be searched and filtered by properties such as article, location, status, review reason, size and Call Off.

Selecting an item opens the same Item Detail modal used from Location Detail.

Push Items are accessed through the Items page using the Push Item filter rather than through a separate page.

Dashboard → Trends

The Trends page uses historical snapshots to show basic changes in warehouse occupancy, total items and reviews over time.

Reviews connect the Locations and Items workflows.

StockScope detects review conditions from the current warehouse data.

A user can inspect the affected location or item and mark a specific review as `Handled`.

OPEN → Mark as Handled → WAITING FOR VERIFICATION

The next successfully confirmed import evaluates the review condition again.

If the problem no longer exists:

WAITING FOR VERIFICATION → RESOLVED

If the problem still exists:

WAITING FOR VERIFICATION → OPEN

A review can also be resolved directly by normal warehouse activity:

OPEN → RESOLVED

Review history remains stored by StockScope regardless of whether the problem was resolved through a user action or through a warehouse data change.

The complete Phase 1 workflow can therefore be summarized as:

Import Data → Dashboard → Locations / Items / Trends → Location or Item Detail → Review Action → Next Import → Review Reconciliation → Snapshot → Updated Dashboard

3. Data Models

4. Services & State

5. Components

UI Design System

StockScope should use a small and consistent design system instead of styling individual components independently.

The design system should eventually define reusable rules for:

- spacing
- typography
- colors
- status indicators
- buttons
- cards
- tables
- location blocks
- warnings
- responsive layout

Reusable Angular components and shared design tokens should be preferred over repeating component-specific CSS.

The specific styling technology does not need to be decided yet.

6. Routing

7. Implementation

8. Validation

9. Testing

10. Data Flow Check

11. Deployment & Production